Chapter 6 — Introduction to EDA

Chapter 5 repaired defects in a table; this chapter asks how to explore that table before formal modeling

Learning objectives

- Define EDA, explain how it differs from confirmatory analysis
- Describe how exploration and cleaning iterate in practice

What EDA does

- EDA combines descriptive statistics, graphical plots, and light transformations
- Typical steps include summarizing variables, visualizing relationships
- The objective is to understand the data itself before choosing models or tests

EDA versus earlier chapters

- Chapter 1 introduced a minimal exploration mindset
- This chapter is the authoritative home for exploration

Example 6.1 — E-commerce customer EDA pass

- Example 6.1 — hands-on module
- Example 6.1 sketches a first pass on a customer table
- Explore the chapter example module
- View files: `modules/chapter6/example1/`

Why EDA matters

- Guides hypothesis formation from patterns you did not expect
- Example 6.4 contrasts linear and nonlinear cues

Example 6.4 — Linear versus nonlinear patterns

- Example 6.4 — hands-on module
- Example 6.4 notes that a strong linear scatter suggests linear regression may fit
- Shape checks from Example 6.2 should precede algorithm selection
- View files: `modules/chapter6/example4/`

Takeaways

- EDA is observation-first exploration before modeling
- It iterates with cleaning
- Summaries plus plots form the core toolkit introduced in the parts that follow

Next

- Complete the quiz for this part
- Continue to data types and measures of central tendency